CN UNIT 1

1-1 DATA COMMUNICATIONS

The term telecommunication means communication at a distance. The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Fundamental Characteristics:

Delivery, Accuracy, Timeliness, Jitter

Topics discussed in this section:

Components (Message, Sender, Receiver, Transmission Medium, Protocol).

Data Représentation (Text, Number, Images, Audio, Video)

Data Flow (Simplex, Duplex, Half Duplex)

1.1

Components

1. **Message.** The **message** is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

2. **Sender.** The **sender** is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

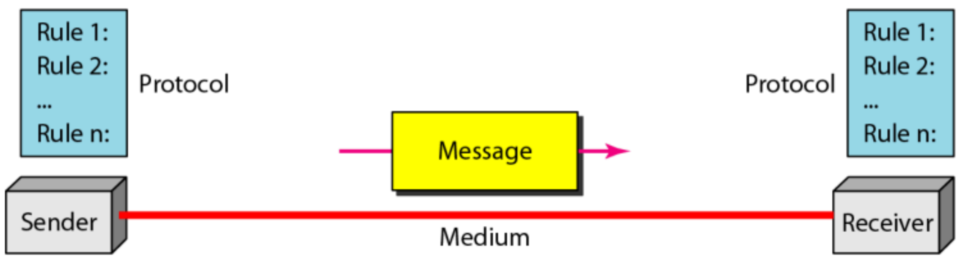
3. **Receiver.** The **receiver** is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

4. **Transmission medium.** The **transmission medium** is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

5. **Protocol.** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

1.1

Figure 1.1 *Five components of data communication*



1.1

Data Flow

- Simplex
- Duplex
  - Half Duplex
  - Full Duplex

1.1

Data Transmission Modes

1. Simplex Mode

- **Definition:** Unidirectional communication (one-way street)
- **How it works:** Only one device transmits; the other only receives
- **Examples:** Keyboards (input only), traditional monitors (output only)
- **Capacity:** Entire channel capacity used for one direction

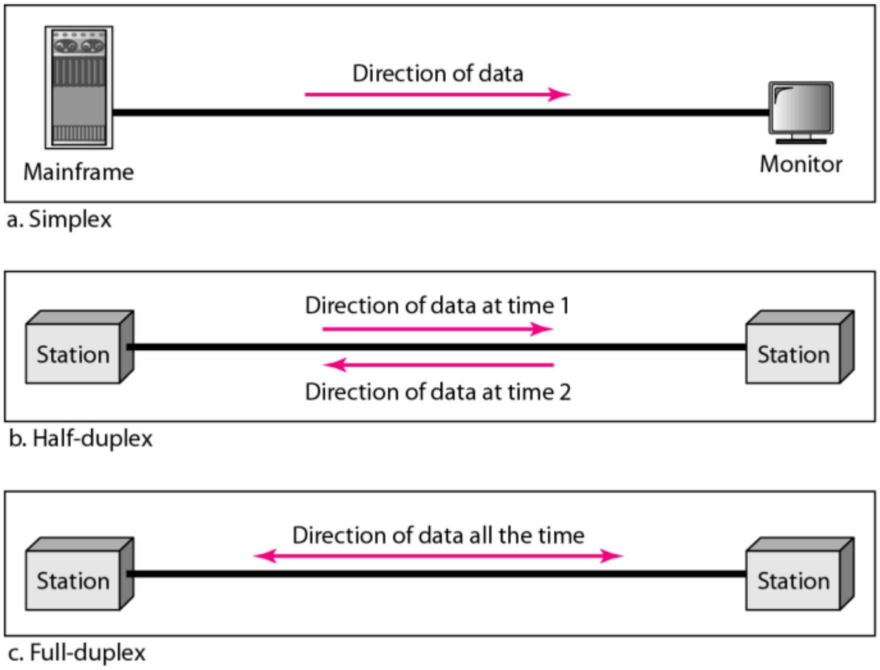
2. Half-Duplex Mode

- **Definition:** Bidirectional communication, but not simultaneously
- **How it works:** Devices can both send and receive, but one at a time
- **Analogy:** One-lane road with traffic in both directions (must take turns)
- **Examples:** Walkie-talkies, CB radios
- **Capacity:** Full channel capacity alternates between directions
- **Use case:** When simultaneous communication isn't needed

3. Full-Duplex Mode

- **Definition:** Bidirectional communication simultaneously (also called duplex)
- **How it works:** Both devices transmit and receive at the same time
- **Analogy:** Two-way street with traffic flowing both directions simultaneously
- **Implementation:**
    - Two physically separate transmission paths, OR
    - Channel capacity divided between both directions
- **Example:** Telephone networks (both people can talk and listen simultaneously)
- **Use case:** When continuous bidirectional communication is required.

Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



1.1

Network Criteria

A network must meet three essential criteria: **performance**, **reliability**, and **security**.

1. Performance

- **Key Metrics:**
  - **Transit time:** Time for a message to travel from one device to another
  - **Response time:** Time between an inquiry and a response
  - **Throughput:** Amount of data transmitted successfully
  - **Delay:** Time lag in data transmission
- **Factors affecting performance:**
  - Number of users
  - Type of transmission medium
  - Hardware capabilities
  - Software efficiency

- **Trade-off:** Higher throughput often leads to increased delay due to traffic congestion

2. Reliability

- **Accuracy:** Correct delivery of data
- **Frequency of failure:** How often the network fails
- **Recovery time:** Time taken to recover from failures
- **Robustness:** Network's ability to withstand catastrophic events

3. Security

- **Protection from unauthorized access:** Prevent intruders from accessing data
- **Protection from damage:** Safeguard data from corruption or loss
- **Recovery procedures:** Policies for handling breaches and data losses

Physical Structures

Type of Connection

A network consists of two or more devices connected through links. A link is the communication pathway that transfers data between devices.

There are two main types of connections:

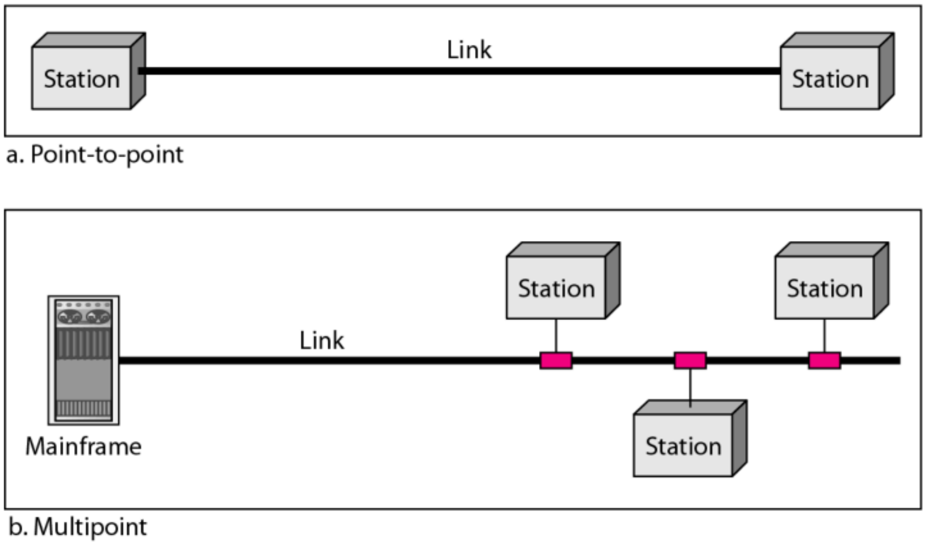
1. Point-to-Point Connection

- Provides a dedicated link between two devices only
- The entire capacity of the link is reserved for transmission between those two devices
- Can use wire, cable, microwave, or satellite links
- **Example:** When you use an infrared remote control to change TV channels, you create a point-to-point connection between the remote and the TV

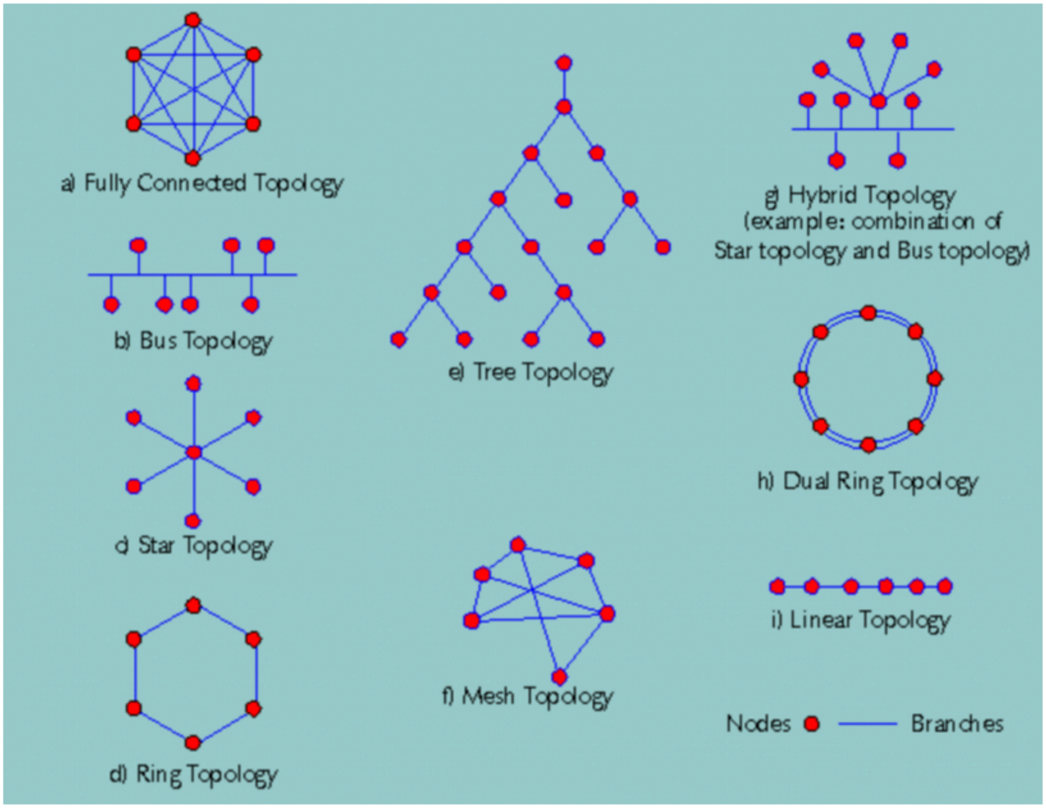
2. Multipoint Connection

- Also called multidrop connection
- More than two devices share a single link
- The channel capacity is shared in two ways:
  - **Spatially shared:** Several devices can use the link at the same time
  - **Timeshared:** Users must take turns using the link

Figure 1.3 *Types of connections: point-to-point and multipoint*



1.1



1.1

Mesh Topology

Main Points

- Every device has a dedicated point-to-point link to every other device
- The term dedicated means the link carries traffic only between the two devices it connects
- Number of physical links required: n(n-1)/2 duplex-mode links for n nodes
- Each device needs n-1 I/O ports to connect to other stations

Advantages

- Dedicated links eliminate traffic problems by allowing each connection to carry its own data load
- Robust topology - if one link fails, it doesn't incapacitate the entire system
- Enhanced privacy and security - messages travel along dedicated lines visible only to intended recipients

- Easy fault identification and isolation - traffic can be routed around suspected problem links

Disadvantages

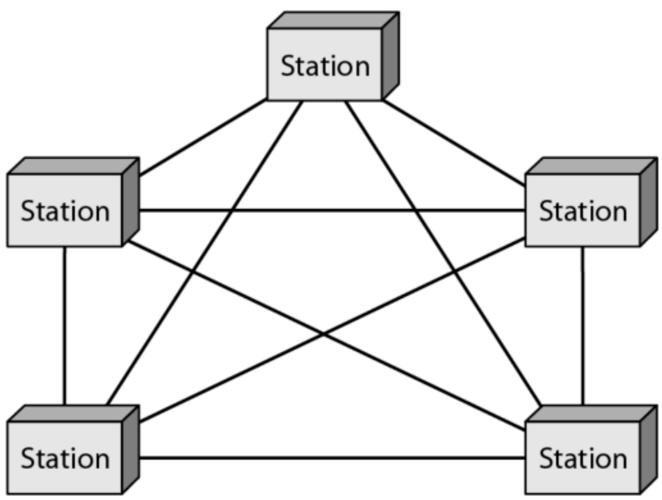
- High cabling requirements - every device must be connected to every other device
- Difficult installation and reconnection
- Large bulk of wiring may exceed available space in walls, ceilings, or floors
- Prohibitively expensive hardware costs for I/O ports and cables
- Usually implemented in limited fashion, such as a backbone in hybrid networks

Practical Example

Connection of telephone regional offices where each regional office needs to be connected to every other regional office

---

Figure 1.5 *A fully connected mesh topology (five devices)*



1.1

Star Topology

Main Points

- Each device has a dedicated point-to-point link only to a central controller (hub)
- Devices are not directly linked to one another
- The hub acts as an exchange, relaying data between devices
- Each device needs only one link and one I/O port

Advantages

- Less expensive than mesh topology
- Easy to install and reconfigure - additions, moves, and deletions involve only one connection
- Less cabling needs to be housed
- Robust - if one link fails, only that link is affected; all other links remain active
- Easy fault identification and isolation - hub can monitor link problems and bypass defective links

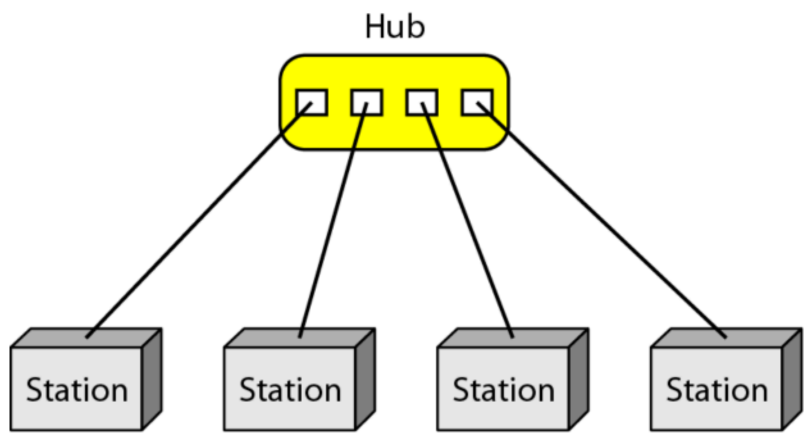
Disadvantages

- Dependency on single point of failure - if the hub goes down, the whole system is dead
- May require more cabling than some other topologies (such as ring or bus) since each node must be linked to central hub

Practical Example

Used in local-area networks (LANs). High-speed LANs often use a star topology with a central hub

Figure 1.6 *A star topology connecting four stations*



1.1

Bus Topology

Main Points

- Multipoint topology where one long cable acts as a backbone to link all devices
- Nodes are connected to the bus cable by drop lines and taps
- Drop line is a connection running between the device and the main cable
- Tap is a connector that splices into or punctures the main cable
- Signal weakens as it travels due to energy transformation into heat
- Limited number of taps and distance between taps

Advantages

- Easy installation - backbone cable can be laid along most efficient path
- Uses less cabling than mesh or star topologies

- No redundancy - only backbone cable stretches through entire facility
- Drop lines only need to reach nearest point on backbone

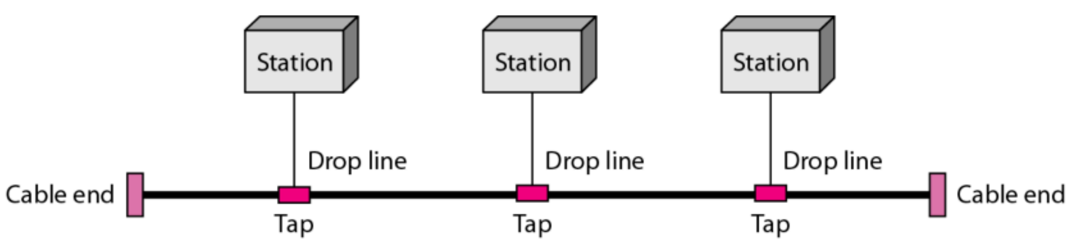
Disadvantages

- Difficult reconnection - usually designed to be optimally efficient at installation
- Difficult to add new devices - may require modification or replacement of backbone
- Signal reflection at taps can cause quality degradation
- Difficult fault isolation
- A fault or break in bus cable stops all transmission, even between devices on same side of problem
- Damaged area reflects signals back, creating noise in both directions

Practical Example

One of the first topologies used in early local-area networks. Ethernet LANs can use a bus topology, but they are less popular now.

Figure 1.7 *A bus topology connecting three stations*



Back Bone Cable are connected to Drop line through Taps.
Signal becomes weak as it travels further distance.
Advantages: Ease of Installation. Uses less cables. Less complex.
Disadvantages: Difficult to fault isolation and reconnection. Difficult to add new devices.
Signal reflection at tapes reduces the signal quality. A fault or break in the bus cable stops all transmission.

1.1

Ring Topology

Main Points

- Each device has a dedicated point-to-point connection with only the two devices on either side
- Signal is passed along the ring in one direction, from device to device, until it reaches destination
- Each device incorporates a repeater that regenerates bits and passes them along

Advantages

- Relatively easy to install and reconfigure
- Each device is linked only to its immediate neighbors
- Adding or deleting a device requires changing only two connections
- Simplified fault isolation - signal circulates at all times, devices can issue alarms if signal not received
- Alerts network operator to problem and its location

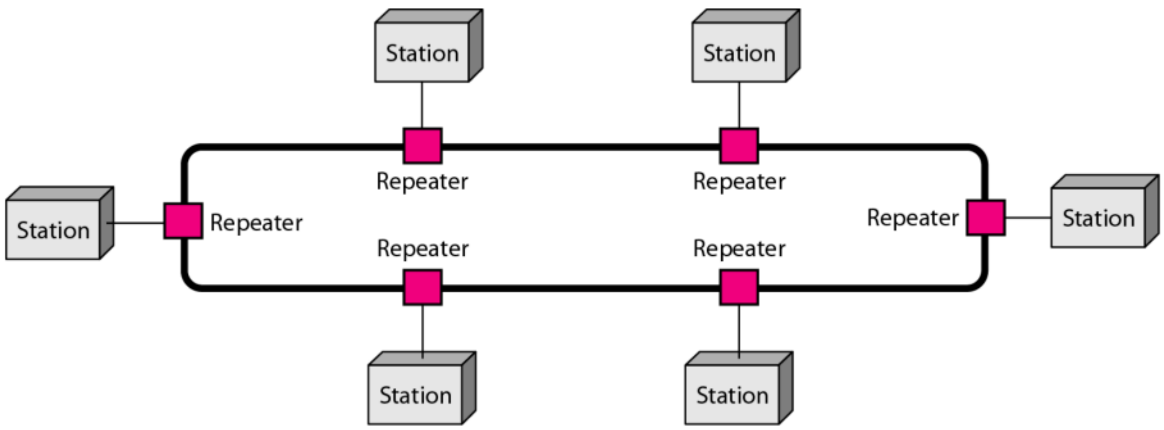
Disadvantages

- Unidirectional traffic can be a disadvantage
- A break in the ring (such as disabled station) can disable entire network
- Weakness can be solved by using dual ring or switch capable of closing off the break

Practical Example

Was prevalent when IBM introduced its local-area network Token Ring. Today, the need for higher-speed LANs has made this topology less popular.

Figure 1.8 *A ring topology connecting six stations*



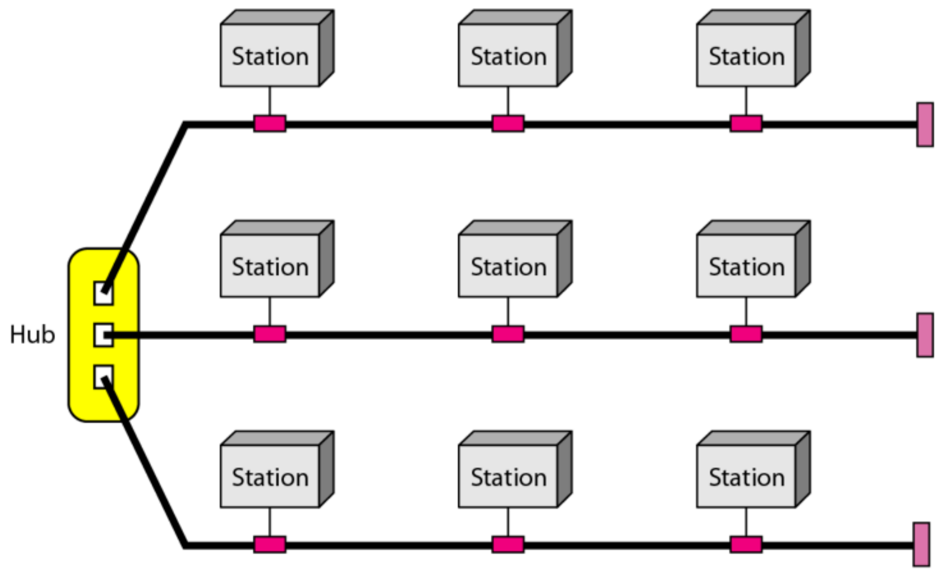
Easy to install and reconfigure. Fault isolation is simplified. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.

The only constraints are media and traffic considerations (maximum ring length and number of devices). Unidirectional traffic can be a disadvantage. In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. Solution: Dual Ring.

1.1

Comparison of Network Topologies

Figure 1.9 *A hybrid topology: a star backbone with three bus networks*



1.1

Topology	Cost	Cabling	Robustness	Fault Isolation	Installation
Mesh	Very High	Extensive	Very High	Easy	Difficult
Star	Moderate	Moderate to High	High (except hub failure)	Easy	Easy
Bus	Low	Low to Moderate	Low	Difficult	Easy initially, difficult to modify
Ring	Moderate	Moderate	Moderate	Easy	Easy

Categories of Networks

Network Classification by Size

- **Local Area Network (LAN):** Covers less than 2 miles
- **Metropolitan Area Network (MAN):** Spans tens of miles (city-sized)

- **Wide Area Network (WAN):** Can be worldwide

Local Area Network (LAN)

Basic Characteristics

- Usually privately owned
- Links devices in a single office, building, or campus
- Size limited to a few kilometers
- Can range from simple (2 PCs and a printer) to complex (entire company with audio/video peripherals)

Purpose

- Allows resource sharing between personal computers or workstations
- Shared resources include:
  - Hardware (e.g., printers)
  - Software (e.g., application programs)
  - Data

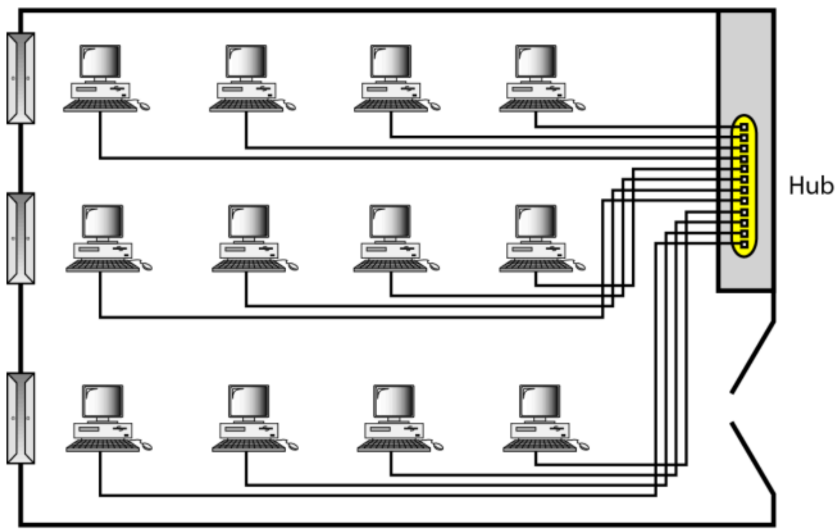
Common Setup

- Links task-related computers (e.g., engineering workstations, accounting PCs)
- One computer acts as a server with large-capacity disk drive
- Software stored on central server, accessible to whole group

Technical Features

- Uses only one type of transmission medium
- Most common topologies: bus, ring, and star
- Early LANs: 4-16 Mbps data rates
- Modern LANs: 100 or 1000 Mbps speeds
- Wireless LANs are the newest evolution
-

Figure 1.10 *An isolated LAN connecting 12 computers to a hub in a closet*



1.1

Wide Area Network (WAN)

Basic Characteristics

- Provides long-distance transmission over large geographic areas
- Can span a country, continent, or the whole world
- Transmits data, images, audio, and video information

Types of WANs

- **Switched WAN:** Complex networks like Internet backbones, connects end systems through routers
- **Point-to-Point WAN:** Simple dial-up line connecting home computer to Internet, leased from telephone or cable TV provider

Common Technologies

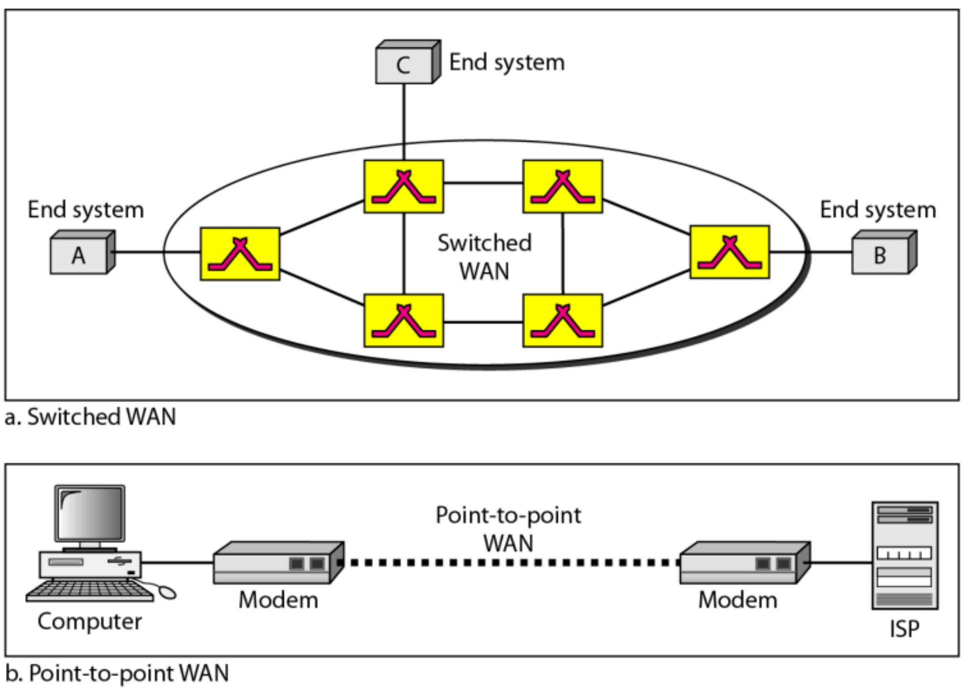
- X.25: Early network for end-user connectivity (being replaced)

- Frame Relay: High-speed, efficient replacement for X.25
- ATM (Asynchronous Transfer Mode): Network using fixed-size data packets called cells
- Wireless WAN: Increasingly popular modern option

Common Uses

- Connecting home computers or small LANs to Internet Service Providers (ISPs)
- Providing Internet access
- Connecting LANs across large distances
-

Figure 1.11 *WANs: a switched WAN and a point-to-point WAN*



1.1

Metropolitan Area Network (MAN)

Basic Characteristics

- Size between LAN and WAN

- Covers area inside a town or city
- Provides high-speed connectivity

Target Users

- Customers needing high-speed connectivity to Internet
- Organizations with endpoints spread across a city or part of a city

Examples

- DSL lines provided by telephone companies
- Cable TV networks used for high-speed Internet connections

2-1 LAYERED TASKS

We use the concept of layers in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.

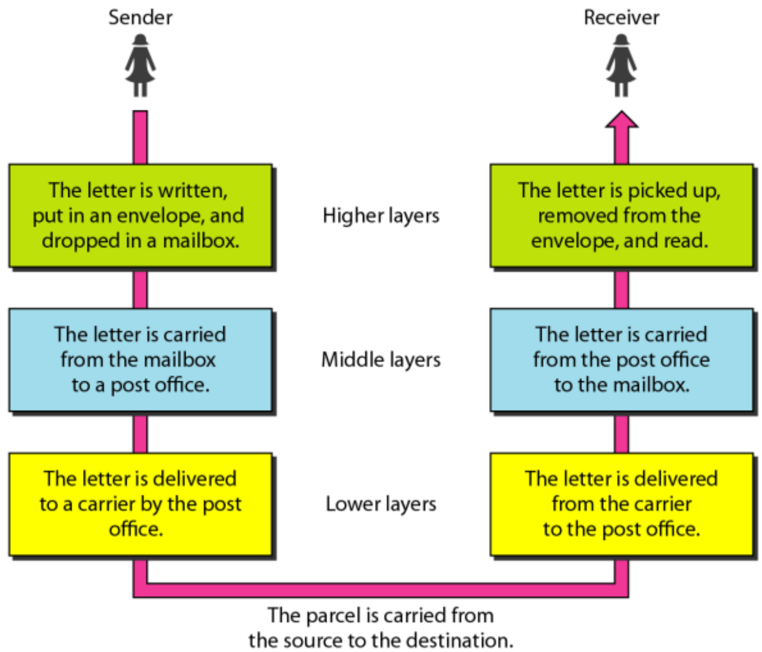
Topics discussed in this section:

Sender, Receiver, and Carrier

Hierarchy

2.1

Figure 2.1 Tasks involved in sending a letter



2.1

2-2 THE OSI MODEL

Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

Topics discussed in this section:

Layered Architecture
Peer-to-Peer Processes
Encapsulation

2.1

Note

ISO is the organization.
OSI is the model.

2.1

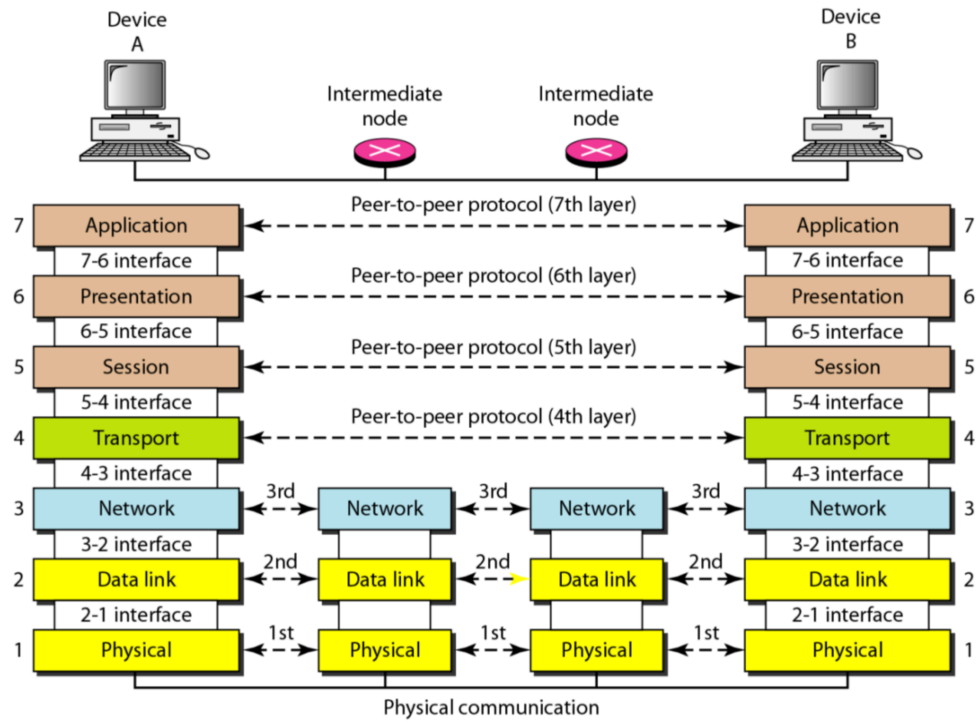
Peer-to-Peer Processes
At the physical layer, communication is direct: In Figure 2.3, device A sends a stream of bits to device B (through intermediate nodes). At the higher layers, however, communication must move down through the layers on device A, over to device B, and then back up through the layers. Each layer in the sending device adds its own information to the message it receives from the layer just above it and passes the whole package to the layer just below it.
At layer I the entire package is converted to a form that can be transmitted to the receiving device. At the receiving machine, the message is unwrapped layer by layer, with each process receiving and removing the data meant for it. For example, layer 2 removes the data meant for it, then passes the rest to layer 3. Layer 3 then removes the data meant for it and passes the rest to layer 4, and so on.

Note

ISO is the organization.
OSI is the model.

2.1

Figure 2.3 *The interaction between layers in the OSI model*



Organization of the Layers

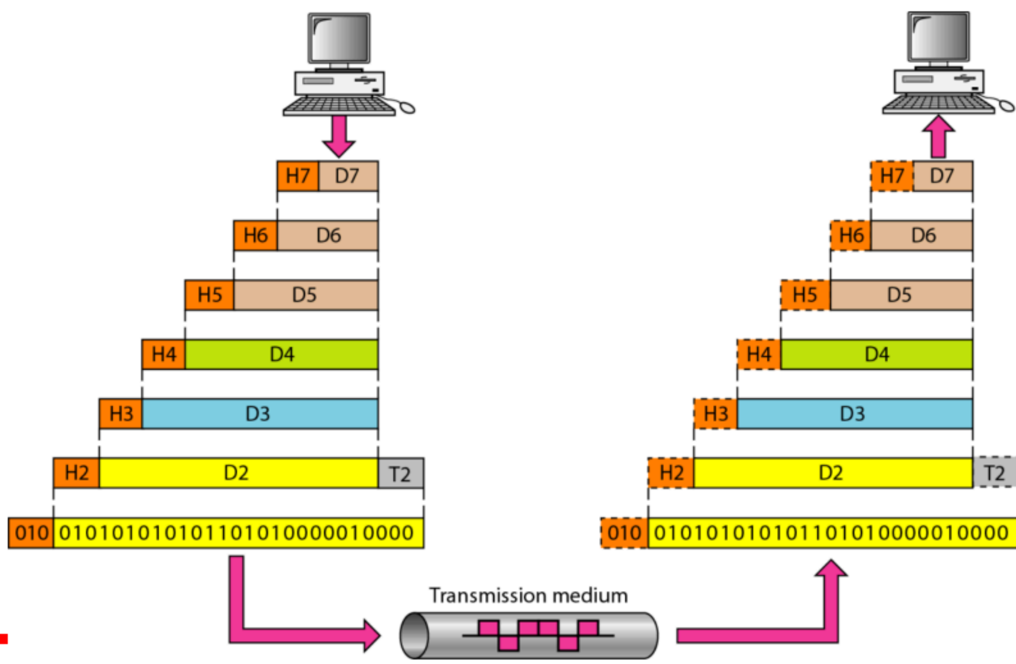
The seven layers can be thought of as belonging to three subgroups. Layers I, 2, and 3-physical, data link, and network-are the network support layers; they deal with the physical aspects of moving data from one device to another (such as electrical specifications, physical connections, physical addressing, and transport timing and reliability). Layers 5, 6, and 7-session, presentation, and application-can be thought of as the user support layers; they allow interoperability among unrelated software systems. Layer 4, the transport layer, links the two subgroups and ensures that what the lower layers have transmitted is in a form that the upper layers can use. The upper OSI layers are almost always implemented in software; lower layers are a combination of hardware and software, except for the physical layer, which is mostly hardware.

In Figure 2.4, which gives an overall view of the OSI layers, D7 means the data unit at layer 7, D6 means the data unit at layer 6, and so on. The process starts at layer 7 (the application layer), then moves from layer to layer in descending, sequential order. At each layer, a header, or possibly a trailer, can be added to the data unit.

Commonly, the trailer is added only at layer 2. When the formatted data unit passes through the physical layer (layer 1), it is changed into an electromagnetic signal and transported along a physical link.

Upon reaching its destination, the signal passes into layer 1 and is transformed back into digital form. The data units then move back up through the OSI layers. As each block of data reaches the next higher layer, the headers and trailers attached to it at the corresponding sending layer are removed, and actions appropriate to that layer are taken. By the time it reaches layer 7, the message is again in a form appropriate to the application and is made available to the recipient.

Figure 2.4 *An exchange using the OSI model*



2.1

Encapsulation

Figure 2.3 reveals another aspect of data communications in the OSI model: encapsulation. A packet (header and data) at level 7 is encapsulated in a packet at level 6. The whole packet at level 6 is encapsulated in a packet at level 5, and so on.
In other words, the data portion of a packet at level N - 1 carries the whole packet (data and header and maybe trailer) from level N. The concept is called encapsulation; level N - 1 is not aware of which part of the encapsulated packet is data and which part

is the header or trailer. For level N - 1, the whole packet coming from level N is treated as one integral unit

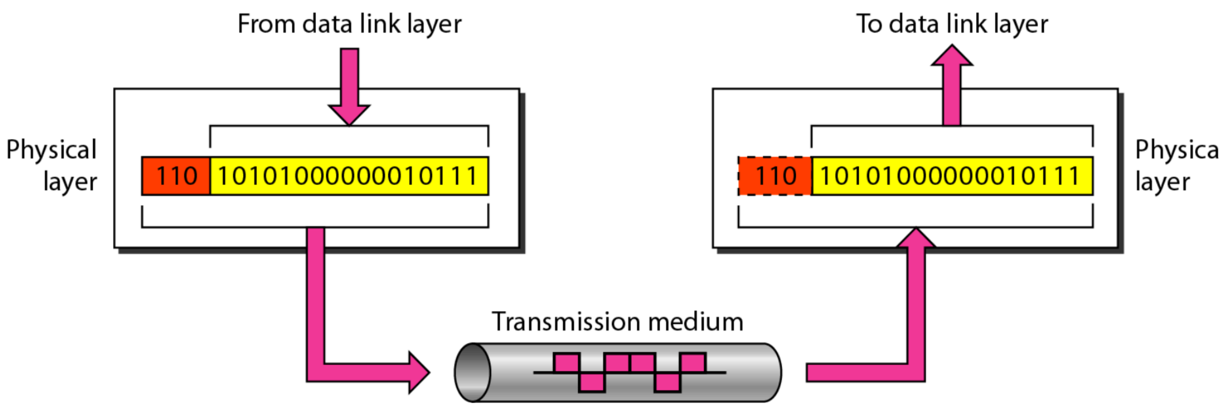
Physical Layer

The physical layer coordinates the functions required to carry a bit stream over a physical medium. It deals with the mechanical and electrical specifications of the interface and transmission medium. It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to Occur. Figure 2.5 shows the position of the physical layer with respect to the transmission medium and the data link layer.

The physical layer is also concerned with the following:
o Physical characteristics of interfaces and medium. The physical layer defines the characteristics of the interface between the devices and the transmission medium. It also defines the type of transmission medium.
o Representation of bits. The physical layer data consists of a stream of bits (sequence of Os or 1s) with no interpretation. To be transmitted, bits must be

encoded into signals--electrical or optical. The physical layer defines the type of encoding (how Os and I s are changed to signals).
o Data rate. The transmission rate-the number of bits sent each second-is also defined by the physical layer. In other words, the physical layer defines the duration of a bit, which is how long it lasts.
o Synchronization of bits. The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level. In other words, the sender and the receiver clocks must be synchronized.
o Line configuration. The physical layer is concerned with the connection of devices to the media. In a point-to-point configuration, two devices are connected through a dedicated link. In a multipoint configuration, a link is shared among several devices.
o Physical topology. The physical topology defines how devices are connected to make a network. Devices can be connected by using a mesh topology (every device is connected to every other device), a star topology (devices are connected through a central device), a ring topology (each device is connected to the next, forming a ring), a bus topology (every device is on a common link), or a hybrid topology (this is a combination of two or more topologies).
o Transmission mode. The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex. In simplex mode, only one device can send; the other can only receive. The simplex mode is a one-way communication. In the half-duplex mode, two devices can send and receive, but not at the same time. In a full-duplex (or simply duplex) mode, two devices can send and receive at the same time.

The physical layer is responsible for movements of individual bits from one hop (node) to the next.



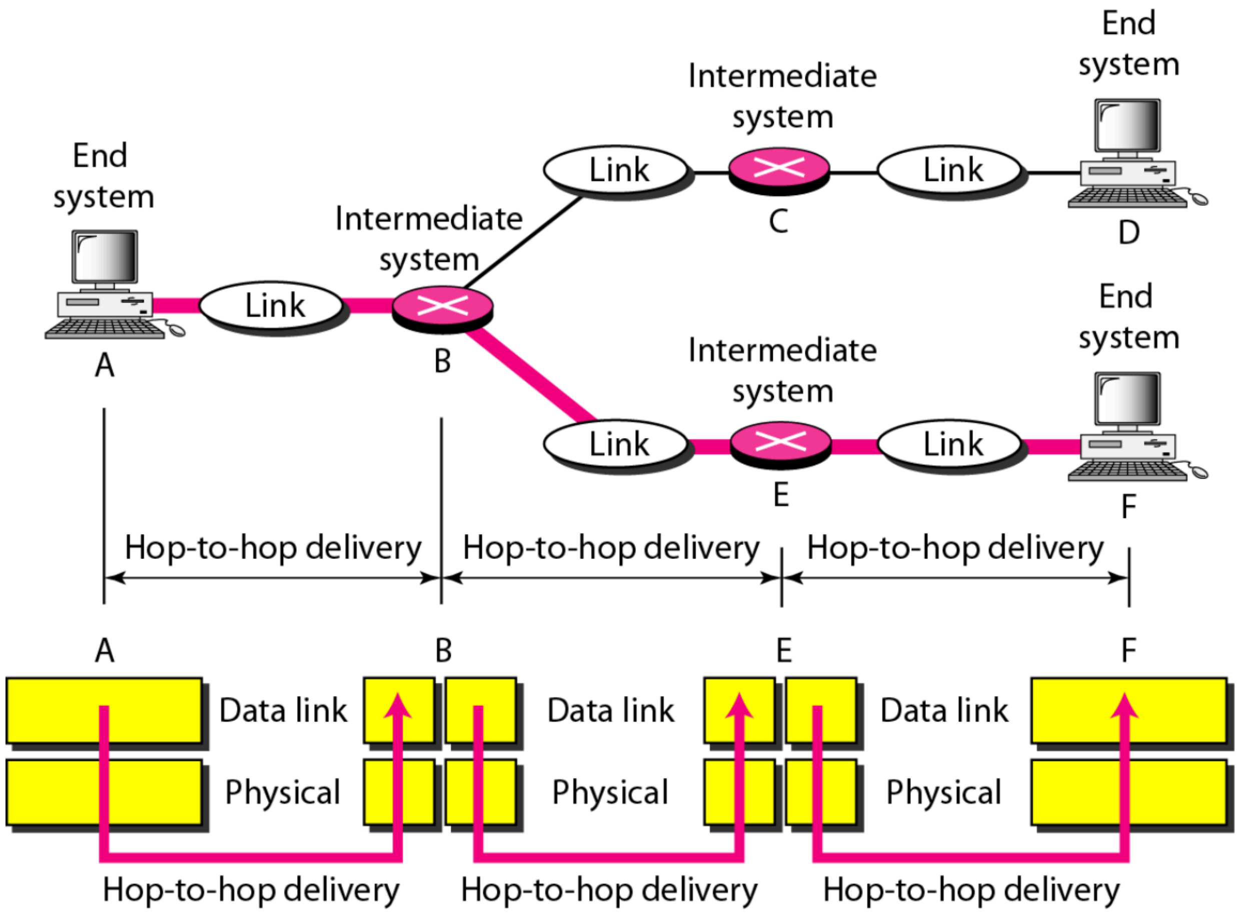
Data Link Layer

The data link layer transforms the physical layer, a raw transmission facility, to a reliable link. It makes the physical layer appear error-free to the upper layer (network layer). Figure 2.6 shows the relationship of the data link layer to the network and physicallayers.

The data link layer is responsible for moving frames from one hop (node) to the next. Other responsibilities of the data link layer include the following:
[I Framing. The data link layer divides the stream of bits received from the network layer into manageable data units called frames.
o Physical addressing. If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame. If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects the network to the next one.
D Flow control. If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.
o Error control. The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplicate frames. Error control is normally achieved through a trailer added to the end of the frame.
D Access control. When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.
Figure 2.7 illustrates hop-to-hop (node-to-node) delivery by the data link layer.

As the figure shows, communication at the data link layer occurs between two adjacent nodes. To send data from A to F, three partial deliveries are made. First, the data link layer at A sends a frame to the data link layer at B (a router). Second, the data

link layer at B sends a new frame to the data link layer at E. Finally, the data link layer at E sends a new frame to the data link layer at F. Note that the frames that are exchanged between the three nodes have different values in the headers. The frame from A to B has B as the destination address and A as the source address. The frame from B to E has E as the destination address and B as the source address. The frame from E to F has F as the destination address and E as the source address. The values of the trailers can also be different if error checking includes the header of the frame.



Network Layer

The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links). Whereas the data link layer oversees the delivery of the packet between two systems on the same network (links), the network layer ensures that each packet gets from its point of origin to its final destination.
If two systems are connected to the same link, there is usually no need for a network layer. However, if the two systems are attached to different networks (links) with connecting devices between the networks (links), there is often a need for the network layer to accomplish source-to-destination delivery. Figure 2.8 shows the relationship of the network layer to the data link and transport layers.

Other responsibilities of the network layer include the following:
o Logical addressing. The physical addressing implemented by the data link layer

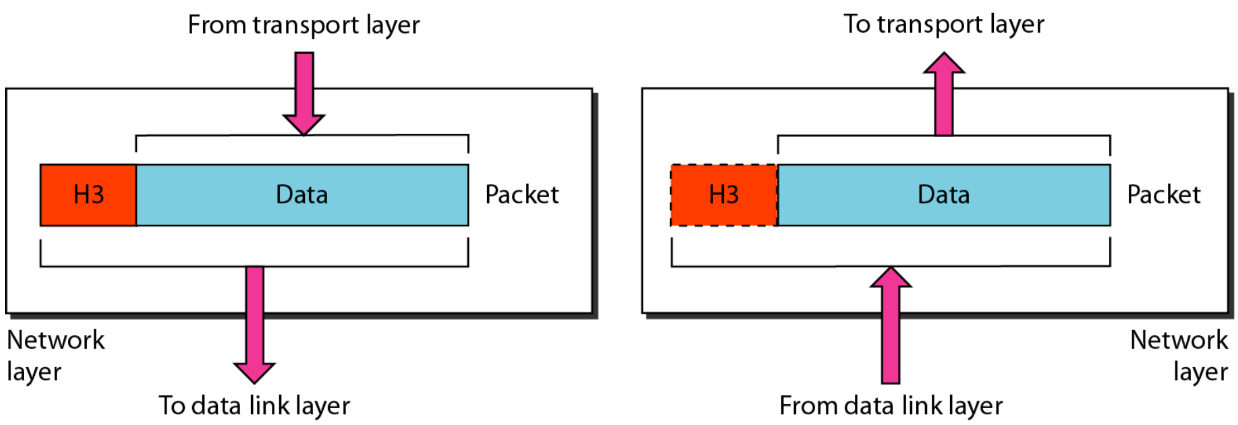
handles the addressing problem locally. If a packet passes the network boundary, we need another addressing system to help distinguish the source and destination systems. The network layer adds a header to the packet coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver. We discuss logical addresses later in this chapter.

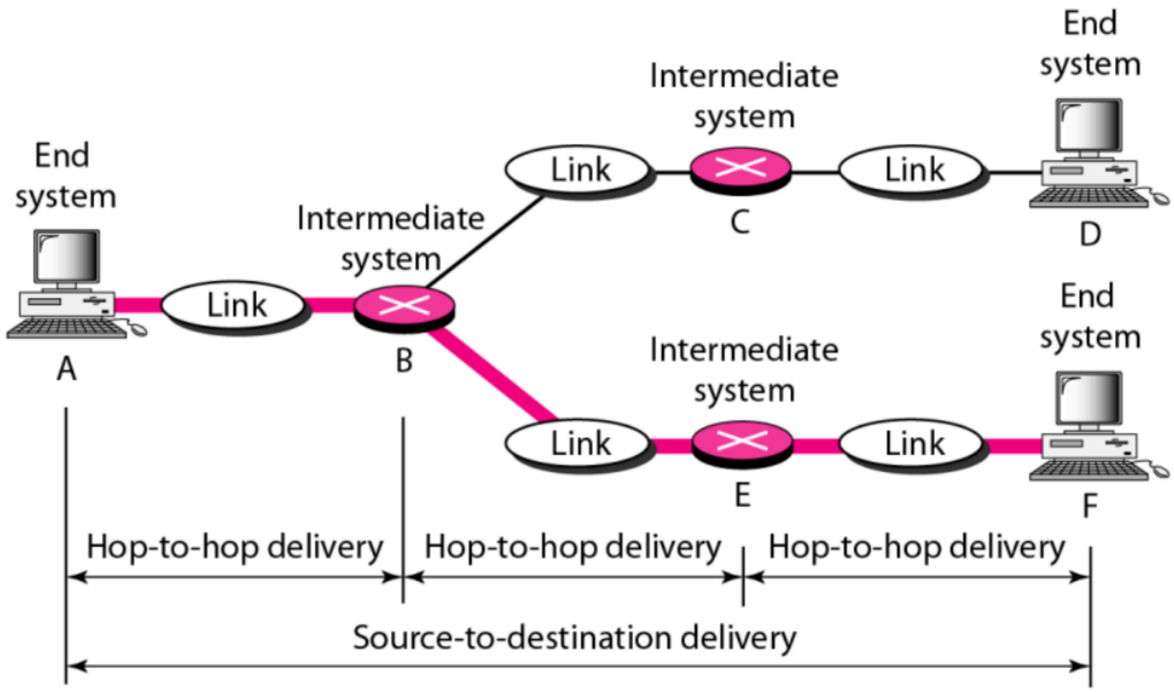
o Routing. When independent networks or links are connected to create intemetworks (network of networks) or a large network, the connecting devices (called routers

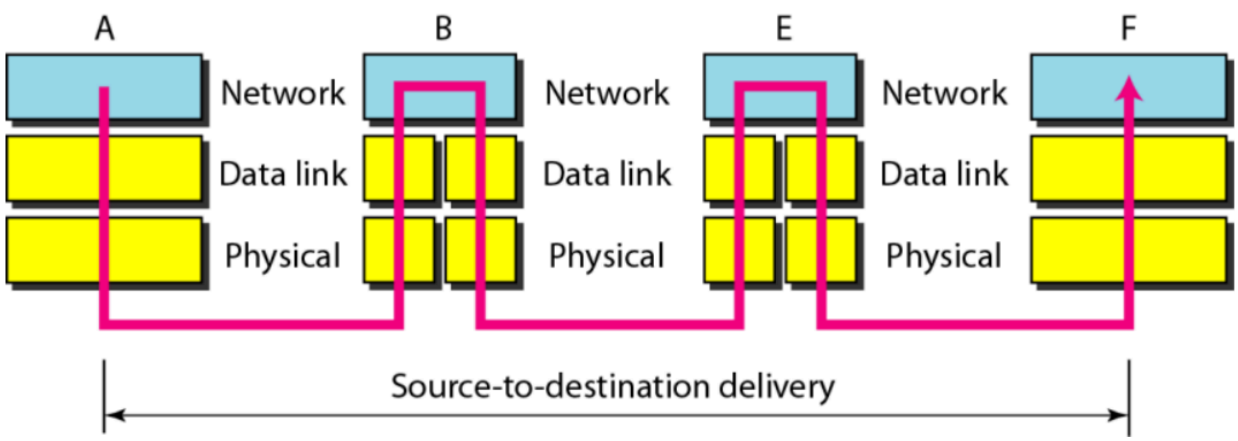
or switches) route or switch the packets to their final destination. One of the functions of the network layer is to provide this mechanism.

Figure 2.9 illustrates end-to-end delivery by the network layer

As the figure shows, now we need a source-to-destination delivery. The network layer at A sends the packet to the network layer at B. When the packet arrives at router B, the router makes a decision based on the final destination (F) of the packet. As we will see in later chapters, router B uses its routing table to find that the next hop is router E. The network layer at B, therefore, sends the packet to the network layer at E. The network layer at E, in tum, sends the packet to the network layer at F.







<u>Transport Layer</u>

The transport layer is responsible for process-to-process delivery of the entire message. A process is an application program running on a host. Whereas the network layer oversees source-to-destination delivery of individual packets, it does not recognize any relationship between those packets. It treats each one independently, as though each piece belonged to a separate message, whether or not it does. The transport layer, on the other hand, ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level. Figure 2.10 shows the relationship of the transport layer to the network and session layers.

Other responsibilities of the transport layer include the following:

o Service-point addressing. Computers often run several programs at the same time. For this reason, source-to-destination delivery means delivery not only from one computer to the next but also from a specific process (running program) on one computer to a specific process (running program) on the other. The transport layer header must therefore include a type of address called a service-point address (or port address). The network layer gets each packet to the correct computer; the transport layer gets the entire message to the correct process on that computer.

o Segmentation and reassembly. A message is divided into transmittable segments, with each segment containing a sequence number. These numbers enable the transport layer to reassemble the message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.

o Connection control. The transport layer can be either connectionless or connectionoriented. A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine. A connectionoriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets. After all the data are transferred, the connection is terminated.

o Flow control. Like the data link layer, the transport layer is responsible for flow control. However, flow control at this layer is performed end to end rather than across a single link.

o Error control. Like the data link layer, the transport layer is responsible for error control. However, error control at this layer is performed process-toprocess rather than across a single link. The sending transport layer makes sure that the entire message arrives at the receiving transport layer without error (damage, loss, or duplication). Error correction is usually achieved through retransmission.

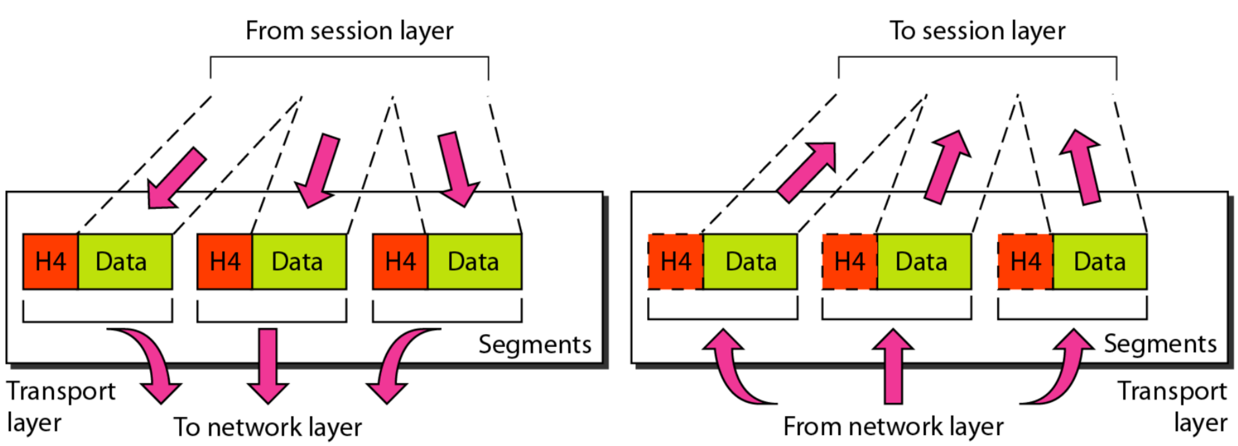
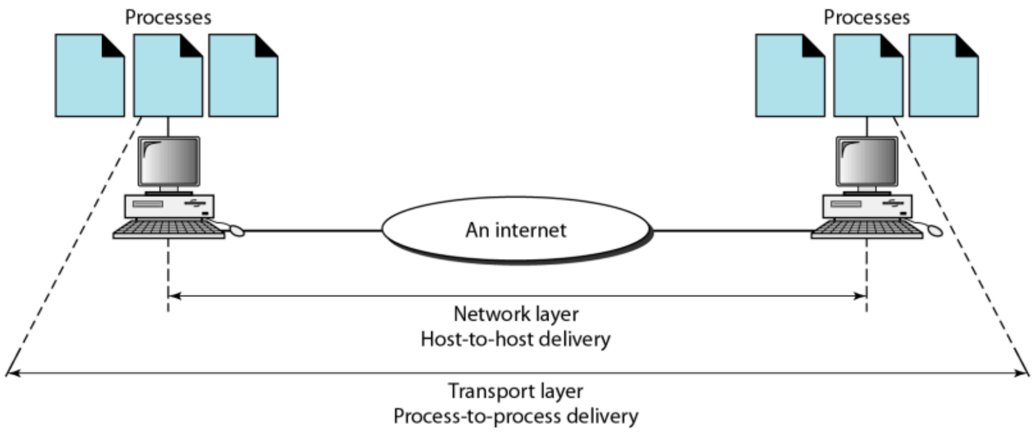


Figure 2.11 *Reliable process-to-process delivery of a message*



2.1

Session Layer

The services provided by the first three layers (physical, data link, and network) are not sufficient for some processes. The session layer is the network dialog controller. It establishes, maintains, and synchronizes the interaction among communicating systems.

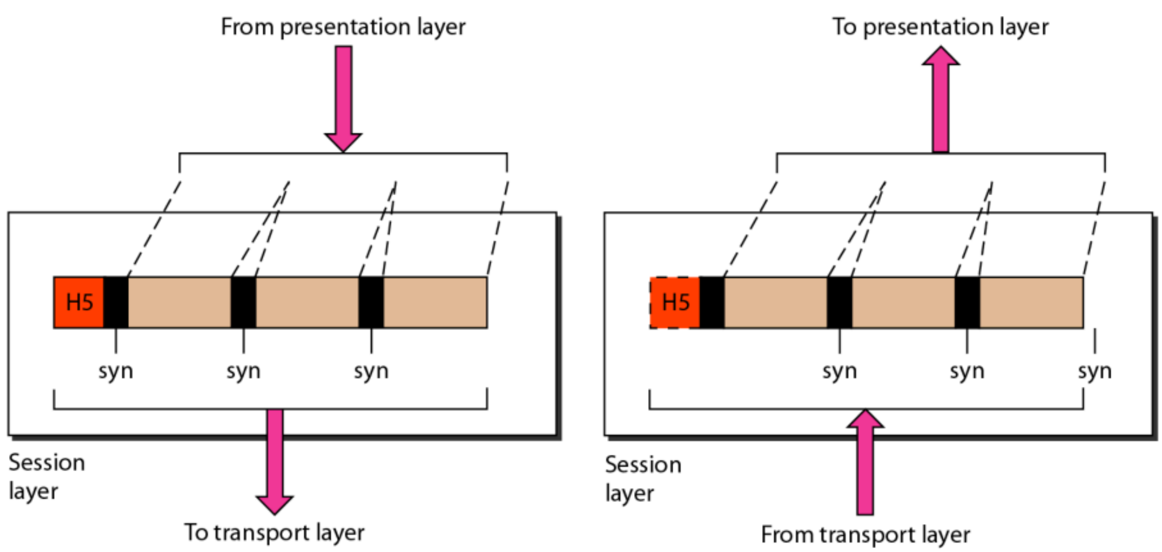
The session layer is responsible for dialog control and synchronization.

Specific responsibilities of the session layer include the following:

o Dialog control. The session layer allows two systems to enter into a dialog. It allows the communication between two processes to take place in either halfduplex (one way at a time) or full-duplex (two ways at a time) mode.

o Synchronization. The session layer allows a process to add checkpoints, or synChronization points, to a stream of data. For example, if a system is sending a file of 2000 pages, it is advisable to insert checkpoints after every 100 pages to ensure that each 100-page unit is received and acknowledged independently. In this case, if a crash happens during the transmission of page 523, the only pages that need to be resent after system recovery are pages 501 to 523. Pages previous to 501 need not be resent. Figure 2.12 illustrates the relationship of the session layer to the transport and presentation layers.

Figure 2.12 *Session layer*



2.1

Presentation Layer

The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems. Figure 2.13 shows the relationship between the presentation layer and the application and session layers.

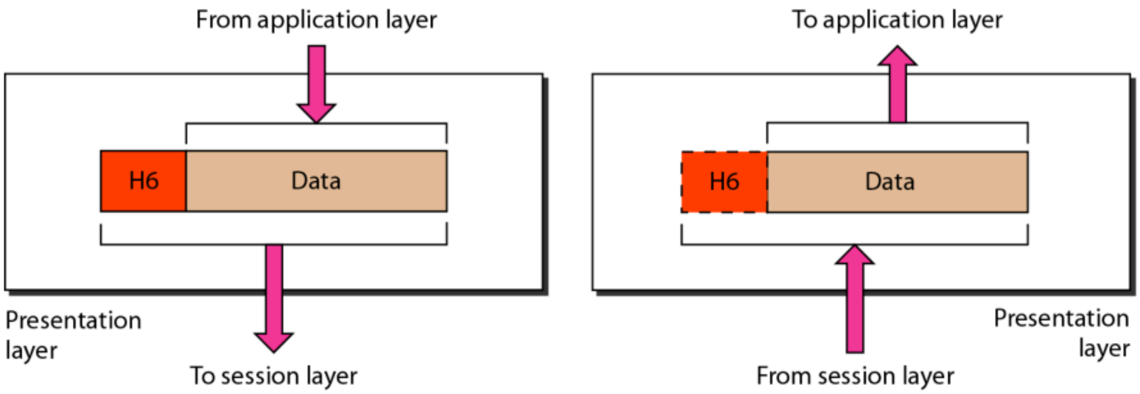
Specific responsibilities of the presentation layer include the following:

o Translation. The processes (running programs) in two systems are usually exchanging information in the form of character strings, numbers, and so on. The infonnation must be changed to bit streams before being transmitted. Because different computers use different encoding systems, the presentation layer is responsible for interoperability between these different encoding methods. The presentation layer at the sender changes the information from its sender-dependent format into a common format. The presentation layer at the receiving machine changes the common format into its receiver-dependent format.

o Encryption. To carry sensitive information, a system must be able to ensure privacy. Encryption means that the sender transforms the original information to another form and sends the resulting message out over the network. Decryption reverses the original process to transform the message back to its original form.

o Compression. Data compression reduces the number of bits contained in the information. Data compression becomes particularly important in the transmission of multimedia such as text, audio, and video.

Figure 2.13 *Presentation layer*



2.1

Application Layer

The application layer enables the user, whether human or software, to access the network. It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.

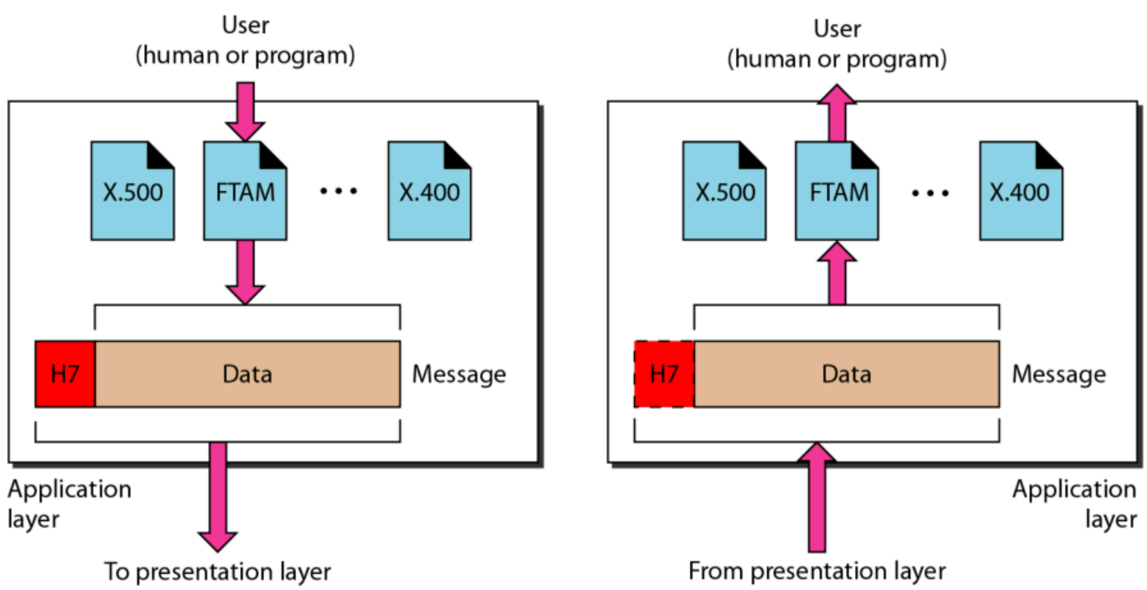
Figure 2.14 shows the relationship of the application layer to the user and the presentation layer. Of the many application services available, the figure shows only three: XAOO (message-handling services), X.500 (directory services), and file transfer, access, and management (FTAM). The user in this example employs XAOO to send an e-mail message.

Specific services provided by the application layer include the following:

o Network virtual terminal. A network virtual terminal is a software version of a physical terminal, and it allows a user to log on to a remote host. To do so, the application creates a software emulation of a terminal at the remote host. The user's computer talks to the software terminal which, in turn, talks to the host, and vice versa. The remote host believes it is communicating with one of its own terminals and allows the user to log on.

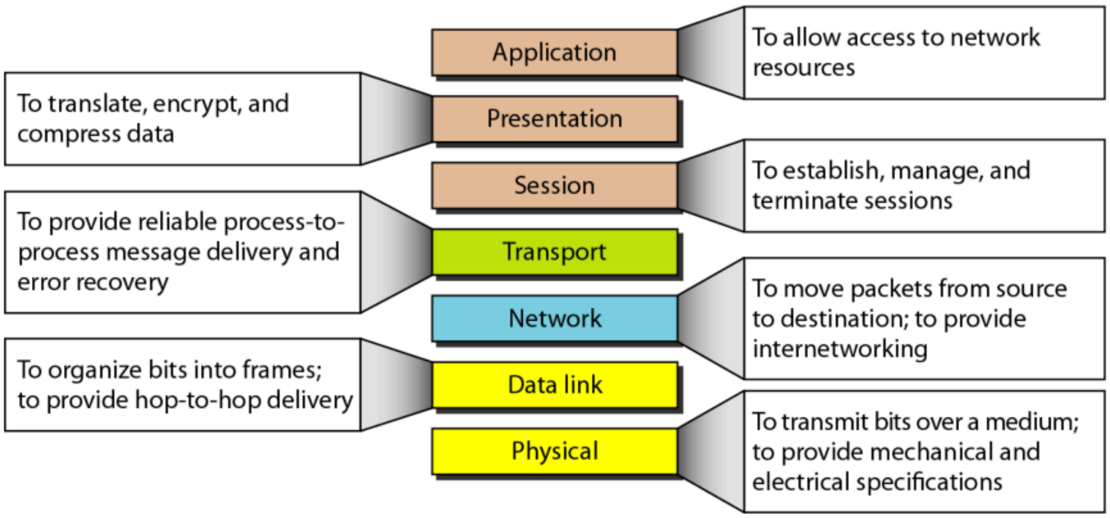
o File transfer, access, and management. This application allows a user to access files in a remote host (to make changes or read data), to retrieve files from a remote computer for use in the local computer, and to manage or control files in a remote computer locally.
o Mail services. This application provides the basis for e-mail forwarding and storage.
o Directory services. This application provides distributed database sources and access for global information about various objects and services.

Figure 2.14 *Application layer*



2.1

Figure 2.15 *Summary of layers*



2.1

TCP/IP PROTOCOL SUITE

Overview

- **Development:** TCP/IP was developed before the OSI model
- **Layer Mismatch:** TCP/IP layers don't exactly match OSI layers
- **Original Structure:** Four layers (host-to-network, internet, transport, application)
- **Modern Structure:** Five layers (physical, data link, network, transport, application)

Layer Comparison with OSI

- **Physical & Data Link:** Equivalent to OSI's physical and data link layers combined
- **Network Layer:** Equivalent to OSI's network layer
- **Transport Layer:** Handles transport + part of session layer duties
- **Application Layer:** Combines OSI's session, presentation, and application layers

Key Characteristics

- **Hierarchical Protocol:** Made of interactive modules with specific functionalities
- **Module Independence:** Modules are not necessarily interdependent
- **Flexibility:** Protocols can be mixed and matched based on system needs
- **Support Structure:** Each upper-level protocol is supported by one or more lower-level protocols

Transport Layer Protocols (3 Main Protocols)

- **TCP** - Transmission Control Protocol
- **UDP** - User Datagram Protocol
- **SCTP** - Stream Control Transmission Protocol

Network Layer Protocols

- **Main Protocol:** IP (Internetworking Protocol)
- **Supporting Protocols:** ARP, RARP, ICMP, IGMP

---

1. PHYSICAL AND DATA LINK LAYERS or HOST NETWORK LAYER

- **Protocol Definition:** TCP/IP does not define specific protocols at these layers
- **Flexibility:** Supports all standard and proprietary protocols
- **Network Types:** Can be LAN (Local Area Network) or WAN (Wide Area Network)

---

2. NETWORK LAYER (INTERNETWORK LAYER)

A. Internetworking Protocol (IP)

- **Function:** Transmission mechanism for TCP/IP protocols
- **Nature:** Unreliable and connectionless protocol
- **Delivery Type:** Best-effort delivery service (no guarantees)
- **Error Handling:** No error checking or tracking
- **Data Units:** Transports data in packets called **datagrams**
- **Routing:** Each datagram transported separately; can travel different routes
- **Arrival:** Datagrams can arrive out of sequence or be duplicated
- **Advantage:** Bare-bones design allows maximum efficiency and customization

B. Address Resolution Protocol (ARP)

- **Purpose:** Associates logical address with physical address
- **Usage:** Finds physical address when Internet address is known
- **Physical Address:** Station address imprinted on NIC (Network Interface Card)

C. Reverse Address Resolution Protocol (RARP)

- **Purpose:** Allows host to discover its Internet address from physical address
- **Use Cases:**
    - Computer connected to network for first time
    - Diskless computer is booted

D. Internet Control Message Protocol (ICMP)

- **Purpose:** Sends notifications of datagram problems back to sender
- **Message Types:**
    - Query messages
    - Error reporting messages

E. Internet Group Message Protocol (IGMP)

- **Purpose:** Facilitates simultaneous message transmission to a group
- **Usage:** Group communication and multicast

---

3. TRANSPORT LAYER

Key Difference from IP: IP is host-to-host (device to device), while transport protocols are process-to-process (program to program)

A. User Datagram Protocol (UDP)

- **Nature:** Simpler of the two standard TCP/IP transport protocols
- **Type:** Process-to-process protocol
- **Features Added:**
    - Port addresses
    - Checksum error control
    - Length information

B. Transmission Control Protocol (TCP)

- **Service Level:** Provides full transport-layer services
- **Type:** Reliable stream transport protocol
- **Connection:** Connection-oriented (connection must be established first)
- **Data Units:** Divides data into smaller units called **segments**
- **Segment Contents:**
  - Sequence number (for reordering)
  - Acknowledgment number (for received segments)
- **Transport:** Segments carried inside IP datagrams
- **Receiving End:** Collects and reorders transmission based on sequence numbers

C. Stream Control Transmission Protocol (SCTP)

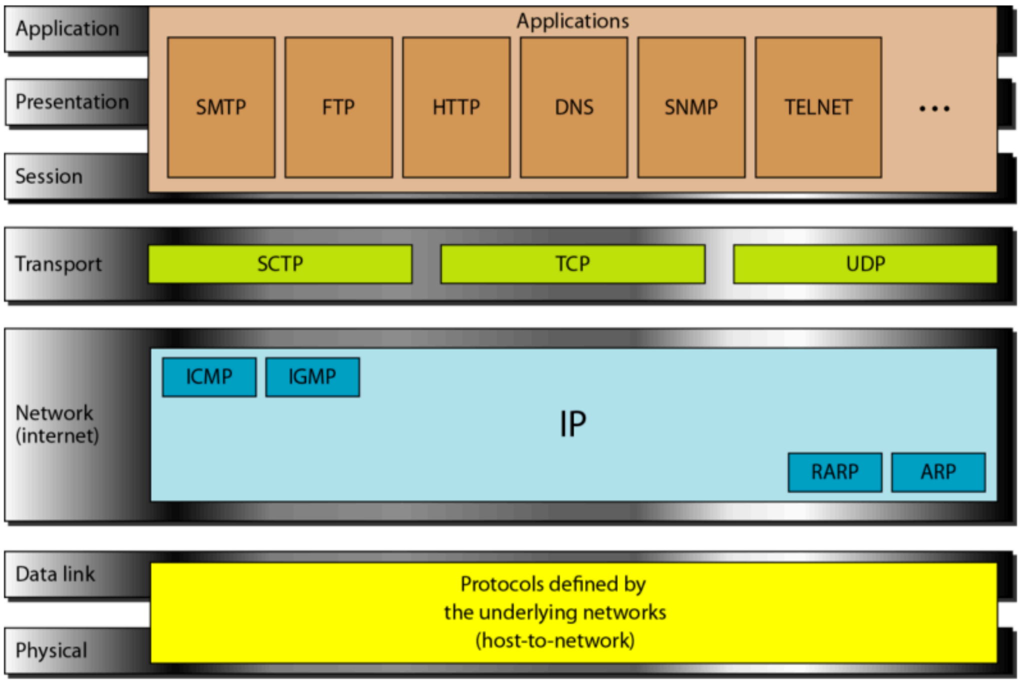
- **Purpose:** Supports newer applications (e.g., voice over Internet)
- **Nature:** Transport layer protocol
- **Design:** Combines best features of UDP and TCP

---

4. APPLICATION LAYER

- **OSI Equivalent:** Combined session, presentation, and application layers
- **Protocols:** Many protocols defined at this layer
- **Coverage:** Standard protocols covered in later chapters

Figure 2.16 *TCP/IP and OSI model*



2.1

2-4 TCP/IP PROTOCOL SUITE

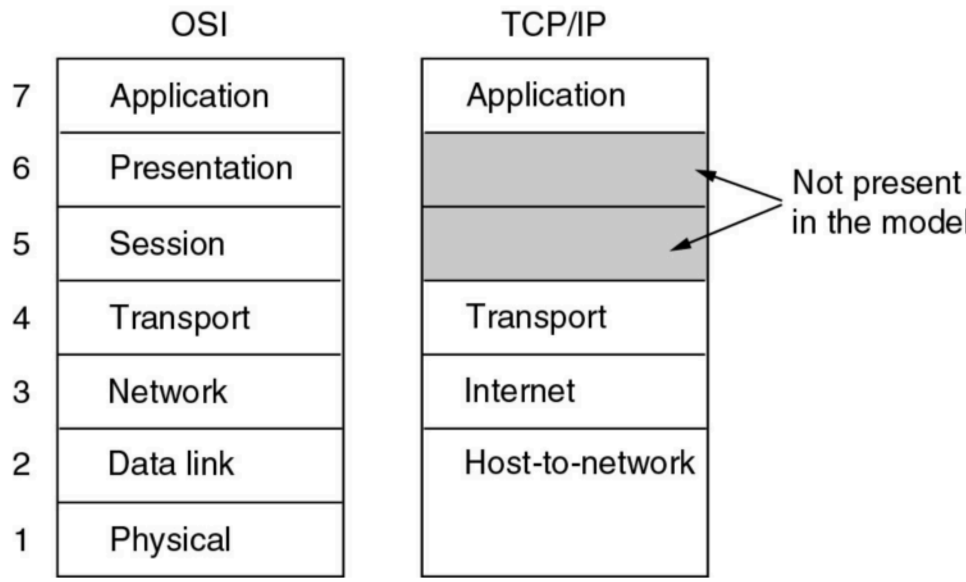
The layers in the TCP/IP protocol suite do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: physical, data link, network, transport, and application.

Topics discussed in this section:

Physical and Data Link Layers
Network Layer
Transport Layer
Application Layer

2.1

The TCP/IP Reference Model



- The TCP/IP reference model.

1

TCP/IP v/s OSI

- 4 Layers
- Did not clearly distinguish between service, interface and protocol.
- Protocols in TCP/IP model are not hidden and tough to replace if technology changes.

- 7 Layers
- Distinction between these three concepts are explicit.
- Protocols in the OSI model are better hidden than in the TCP/IP model and can be replaced relatively easily as the technology changes.

1

- The protocols came first, and the model was really just a description of the existing protocols.
- Designers have much experience with the subject and have clear idea of which functionality to put in which layer.

- The model was not biased toward one particular set of protocols, a fact that made it quite general.
- Designers did not have much experience with the subject and did not have a good idea of which functionality to put in which layer.

1

2.5 ADDRESSING

Four levels of addresses are used in an internet employing the TCP/IP protocols: physical (link) addresses, logical (IP) addresses, port addresses, and specific addresses.

---

1. Physical Addresses

Definition: The physical address (link address) is the address of a node as defined by its LAN or WAN, included in the frame used by the data link layer. It is the lowest-level address.

Characteristics:

- Size and format vary depending on the network
- Ethernet uses a 6-byte (48-bit) address imprinted on the NIC
- LocalTalk (Apple) uses a 1-byte dynamic address that changes at startup

Example 2.1: Physical Address Communication

Scenario: Node with physical address 10 sends a frame to node with physical address 87 via a bus topology LAN.

Process:

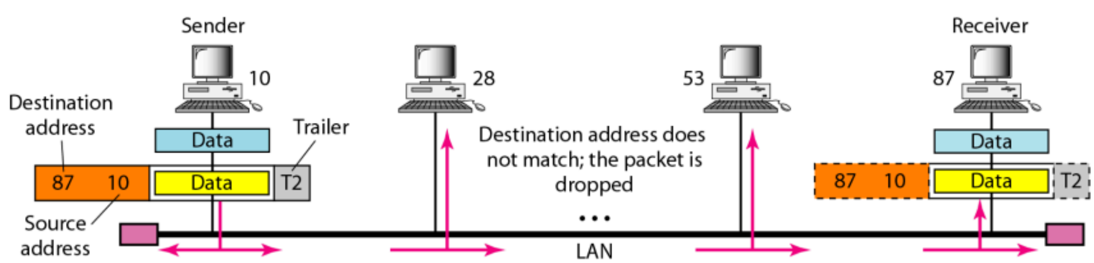
- Data link layer encapsulates data in a frame with header and trailer
- Header contains receiver (87) and sender (10) physical addresses
- Destination address typically comes before source address
- Frame propagates in both directions on the bus
- Each station checks the destination address
- Only the matching station (87) accepts the frame; others drop it
- Frame is decapsulated and data delivered to upper layer

Example 2.2: Physical Address Format

Most LANs use a 48-bit (6-byte) physical address written as 12 hexadecimal digits, with each byte separated by a colon:

`07:01:02:01:2C:4B`

Figure 2.19 *Physical addresses*



2.1

2. Logical Addresses

Purpose: Logical addresses enable universal communications independent of underlying physical networks. They provide unique identification for each host on the Internet.

Key Features:

- Currently a 32-bit address in the Internet
- Uniquely defines a host connected to the Internet
- No two publicly addressed hosts can have the same IP address

Example 2.3: Logical Address Routing

Scenario: An internet with two routers connecting three LANs. Computer with logical address A and physical address 10 sends a packet to computer with logical address P and physical address 95.

Address Requirements:

- Each device has a pair of addresses (logical and physical) for each connection
- Routers have multiple pairs of addresses (one for each network connection)

Transmission Process:

1. Sender encapsulates data in a packet with logical addresses (source A, destination P)
2. Network layer consults routing table to find next hop (router 1, logical address F)
3. ARP finds physical address (20) corresponding to logical address F
4. Data link layer encapsulates packet with physical addresses (source 10, destination 20)
5. Router 1 receives frame, decapsulates packet, reads logical destination P
6. Router 1 forwards packet, creating new frame with updated physical addresses (source 99, destination 33)
7. Router 2 repeats the process, forwarding to final destination
8. At destination, packet is decapsulated and data delivered to upper layer

> **Key Principle:** Physical addresses change from hop to hop, but logical addresses remain the same from source to destination.

---

3. Port Addresses

Purpose: Port addresses enable process-to-process communication. They label different processes running on the same computer to ensure data reaches the correct application.

Characteristics:

- 16 bits in length in TCP/IP architecture
- Allow multiple processes to communicate simultaneously
- Enable end-to-end process communication across the Internet

Example 2.4: Port Address Communication

Scenario: Sending computer runs three processes (port addresses a, b, c). Receiving computer runs two processes (port addresses j, k). Process a needs to communicate with process j.

Encapsulation Process:

1. Transport layer encapsulates data with port addresses (source a, destination j)
2. Network layer encapsulates with logical addresses (source A, destination P)
3. Data link layer encapsulates with physical addresses (change at each hop)

> **Key Principle:** Physical addresses change from hop to hop, but logical and port addresses remain the same from source to destination.

Example 2.5: Port Address Format

A port address is a 16-bit address represented by one decimal number:
`753`

---

4. Specific Addresses

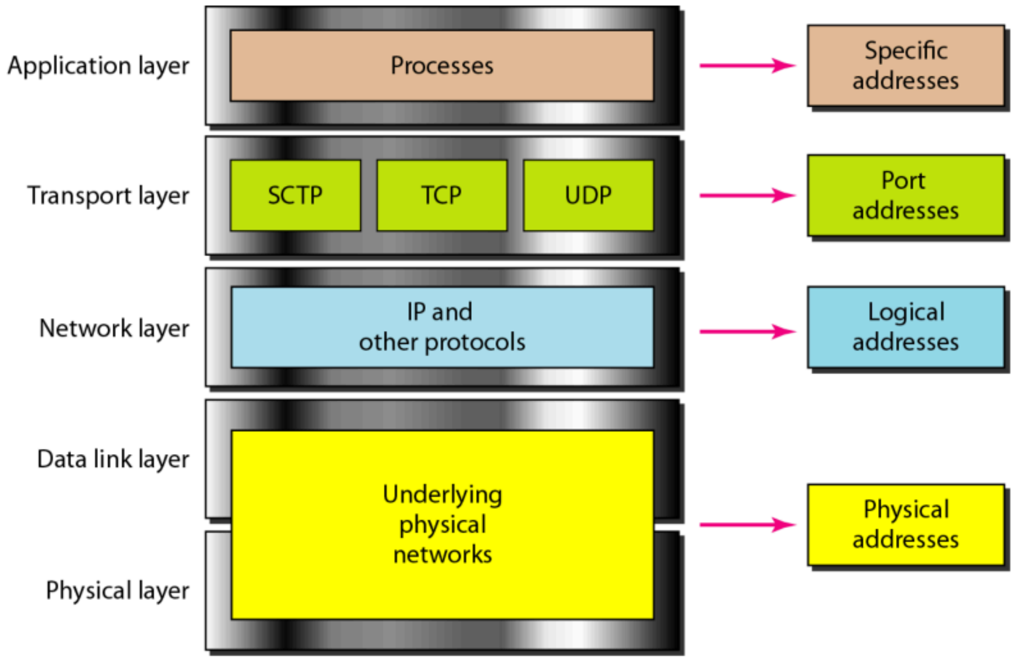
Purpose: User-friendly addresses designed for specific applications.

Examples:

- **E-mail address:** forouzan@fhda.edu (defines recipient of e-mail)
- **URL:** www.mhhe.com (locates documents on the World Wide Web)

Note: These addresses are converted to corresponding port and logical addresses by the sending computer.

Figure 2.18 *Relationship of layers and addresses in TCP/IP*



2.1

The physical addresses change from hop to hop,
but the logical and port addresses usually remain the same

3-4 TRANSMISSION IMPAIRMENT

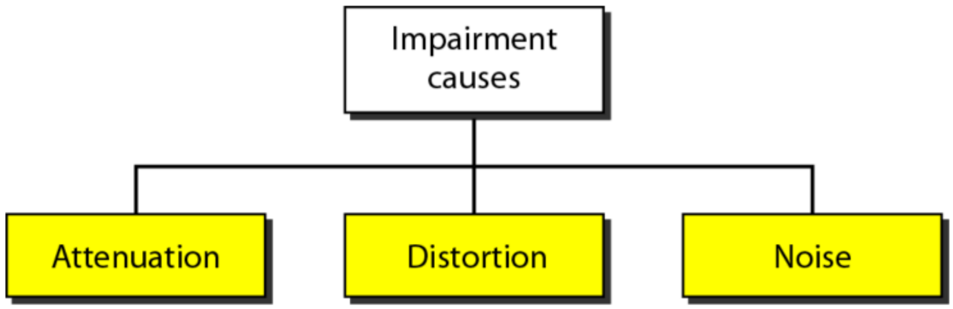
Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise.

Topics discussed in this section:

Attenuation
Distortion
Noise

3.1

Figure 3.25 Causes of impairment



3.1

Attenuation

- Attenuation means a loss of energy.
- When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.
- To show that a signal has lost or gained strength, engineers use the unit of the decibel.
- The decibel (dB) measures the relative strengths of two signals or one signal at two different points.

3.1

Distortion

- Distortion means that the signal changes its form or shape.
- Distortion can occur in a composite signal made of different frequencies.
- As a result, signal components at the receiver have phases different from what they had at the sender.
- The shape of the composite signal is therefore not the same.

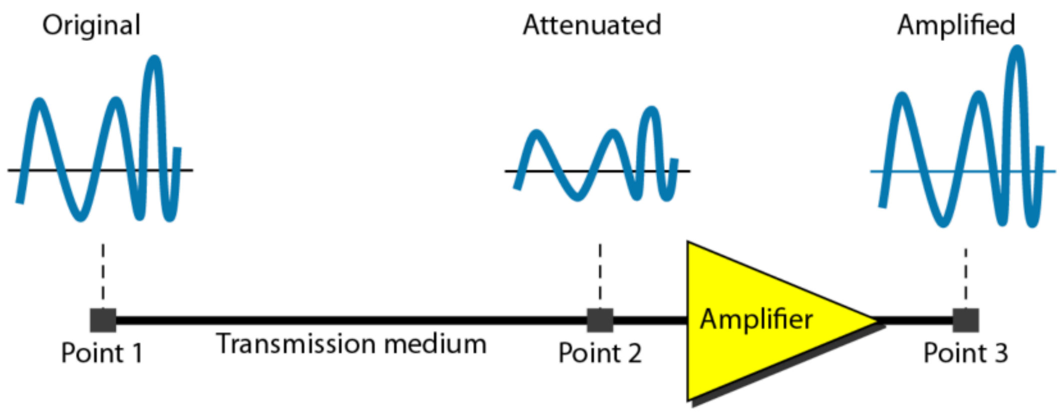
3.1

Noise

- Noise is another cause of impairment.
- Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.
- **Thermal noise** is the random motion of electrons in a wire, which creates an extra signal not originally sent by the transmitter.
- **Induced noise** comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna.
- **Crosstalk** is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna.
- **Impulse noise** is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.

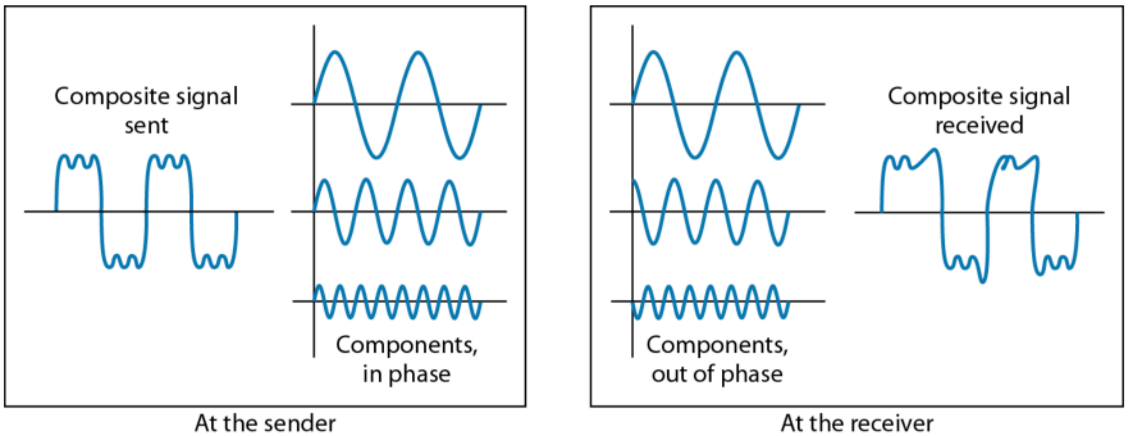
3.1

Figure 3.26 Attenuation



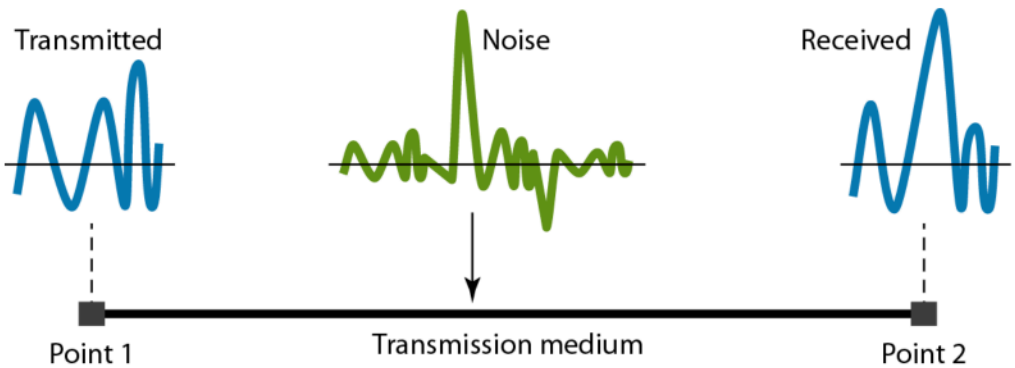
3.1

Figure 3.28 Distortion



3.1

Figure 3.29 Noise



3.1

3-5 DATA RATE LIMITS

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

1. ***The bandwidth available***
2. ***The level of the signals we use***
3. ***The quality of the channel (the level of noise)***

Topics discussed in this section:

Noiseless Channel: Nyquist Bit Rate
Noisy Channel: Shannon Capacity
Using Both Limits

3.1

Data Rate Calculation

Nyquist

Noiseless Channel

L = signal levels

$$BitRate = 2 \times bandwidth \times \log_2 L$$

Shannon

Noisy Channel

SNR = signal to noise ratio

$$Capacity = bandwidth \times \log_2(1 + SNR)$$

3.1

3-6 PERFORMANCE

One important issue in networking is the performance of the network—how good is it? We discuss quality of service, an overall measurement of network performance, in greater detail in Chapter 24. In this section, we introduce terms that we need for future chapters.

Topics discussed in this section:

Bandwidth
Throughput
Latency (Delay)
Bandwidth-Delay Product

3.1

Performance

- Bandwidth: a range of frequencies within a given band, in particular that used for transmitting a signal.
- Throughput: the amount of bits passing through a system or process at given time.
- Latency (Delay): the total time it takes a data packet to travel from one node to another.
- Bandwidth-Delay Product: the product of a data link's capacity (in bits per second) and its round-trip delay time (in seconds).

3.1

Note

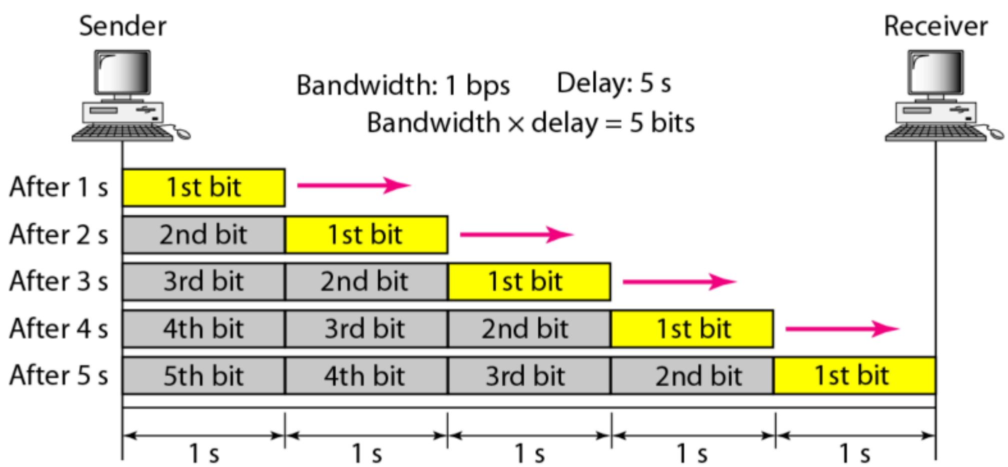
In networking, we use the term bandwidth in two contexts.

- The first, bandwidth in hertz, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.

- The second, bandwidth in bits per second, refers to the speed of bit transmission in a channel or link.

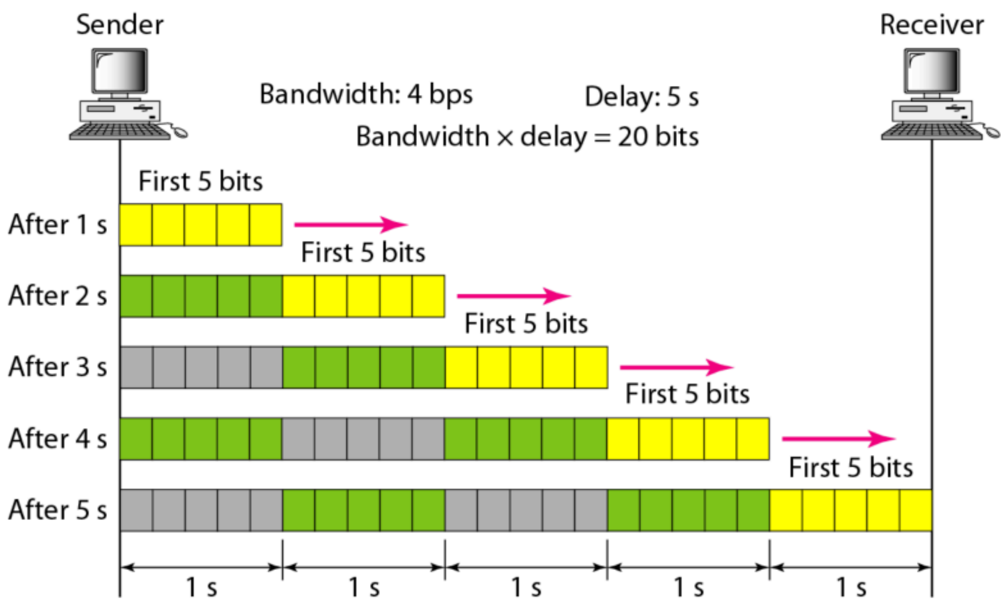
3.1

Figure 3.31 Filling the link with bits for case 1 Frequency Based. FDM



3.1

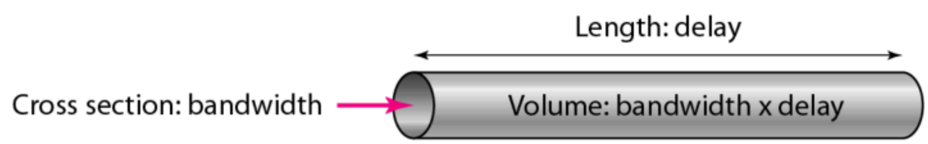
Figure 3.32 Filling the link with bits in case 2 Time Based. TDM



3.1

The bandwidth-delay product defines the number of bits that can fill the link.

Figure 3.33 Concept of bandwidth-delay product



3.1

NYQUISTE=Increasing the levels of a signal may reduce the reliability of the system.